


POLITÉCNICA

# EXAMEN PARCIAL FICHEROS CURSO 2024/2025 (OCTUBRE)

**Fernando Pérez Costoya**

**Curso 2026/27**

# ENUNCIADO

datsi © UPM 2024

POLITÉCNICA

- Bloques de 1 KiB y agrupaciones de 4 bloques.
- i-nodos ocupados del 2 al 53, el resto libres.
- Agrupaciones ocupadas de la 1 a la 89, el resto libres.
- La asignación de agrupaciones e i-nodos se hace seleccionando el primer recurso disponible
- En la tabla intermedia, las entradas 1-10 están ocupadas. A partir de la 11, libres.
- El directorio `/home/usuario1/` existe y tiene: UID = 101, GID = 23, y permisos = 0750.

- **3 procesos de ese usuario (UID=101; GID=23): PA, PB (hijo de PA) y PC**
- **Fichero *fichero1.txt* no existe previamente**

**Se pide**

**a) Tabla en memoria en puntos de control**

**b) Permisos de *fichero1.txt***

**c) Contenido del fichero al final**

**d) Valor variables *n* y *buffer* después de ejecutar línea 14**

**e) Valor variables *n* y *buffer* después de ejecutar línea 16**

# ENUNCIADO



POLITÉCNICA

| Proceso | Código |
|---|---|
| PA | 1: umask(0022);<br>2: fd=creat("/home/usuario1/fichero1.txt", 0666);<br>3: for (i=0;i<1000;i++)<br>4: n=write(fd, "AAAAA", 5);<br>5: pid=fork(); // creación de proceso hijo PB |
| Punto de control 1 | |
| PB | 6: lseek(fd, -2000, SEEK_END);<br>7: n=write(fd, "BBBBB", 5); |
| Punto de control 2 | |
| PC | 8: fd=open("/home/usuario1/fichero1.txt", O_RDWR);<br>9: lseek(fd, 2000, SEEK_CUR);<br>10: n=write(fd, "CCCCC", 5); |
| Punto de control 3 | |
| PB | 11: lseek(fd, 1000, SEEK_SET);<br>12: n=write(fd, "DDDDD", 5); |
| Punto de control 4 | |
| PA | 13: lseek(fd, 1000, SEEK_SET);<br>14: n=read(fd, buffer, 5); |
| PC | 15: lseek(fd, 1000, SEEK_SET);<br>16: n=read(fd, buffer, 5); |

# APARTADO A: ESTADO INICIAL

datsi © UPM 2024

POLITÉCNICA

## Tablas en memoria

**BCP PA**

fd

| | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 0 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask=

???

UID=101

GID=23

**BCP PC**

fd

| | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 0 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask=

???

UID=101

GID=23

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | | | | |
| 12 | | | | |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

**Tamaño de agrupación = 4 * tamaño del bloque = 4KiB**

# APARTADO A: PUNTO DE CONTROL 1; PASO 1

datsi © UPM 2024

## Tablas en memoria

**BCP PA**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022

UID=101

GID=23

**BCP PC**

| fd | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 0 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= ???

UID=101

GID=23

**Tabla intermedia**

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | 0 | 1 | 01 |
| 12 | | | | |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

**Tabla de inodos en memoria**

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 1 | - | 0 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

creat 0666 umask 0022

permisos 0644

**SOLUCIÓN APARTADO B**

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=0 agrup=-

**PA**

```
1: umask(0022);
2: fd=creat("/home/usuario1/fichero1.txt", 0666);
```

devuelve fd=3

# APARTADO A: PUNTO DE CONTROL 1; PASO 2

datsi © UPM 2024

POLITÉCNICA

## Tablas en memoria

**BCP PA**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PC**

| fd | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 0 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= ???
UID=101
GID=23

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | ~~0~~ 5000 | 1 | 01 |
| 12 | | | | |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 1 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA.....AAA

PA

```
3: for (i=0;i<1000;i++)
4:   n=write(fd, "AAAAA", 5);
```

fd=3

datsi © UPM 2024

POLITÉCNICA

## Tablas en memoria

| BCP PA | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PB | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PC | |
|---|---|
| fd | |
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 0 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| ??? | |
| UID=101 | |
| GID=23 | |

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | 5000 | 2 | 01 |
| 12 | | | | |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 1 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA.....AAA

PA

```
5: pid=fork(); // creación de proceso hijo PB
```

# APARTADO A: PUNTO DE CONTROL 2; PASO 1

datsi © UPM 2024

## Tablas en memoria

| BCP PA | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PB | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PC | |
|---|---|
| fd | |
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 0 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| ??? | |
| UID=101 | |
| GID=23 | |

11

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| | 54 | ~~5000~~ 3000 | 2 | 01 |
| 12 | | | | |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 1 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

posición 3000

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA……AAA

PB

```
6: lseek(fd, -2000, SEEK_END);
```

fd=3

datsi © UPM 2024

POLITÉCNICA

## Tablas en memoria

**BCP PA**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PB**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PC**

| fd | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 0 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= ???
UID=101
GID=23

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | ~~3000~~-3005 | 2 | 01 |
| 12 | | | | |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 1 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

posición 3000

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA……AAABBBBBAAA…AAA

**PB**

```
7: n=write(fd, "BBBBB", 5);
```

fd=3

# APARTADO A: PUNTO DE CONTROL 3; PASO1

datsi © UPM 2024

## Tablas en memoria

**BCP PA**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PB**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PC**

| fd | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 12 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= ???
UID=101
GID=23

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | 3005 | 2 | 01 |
| 12 | 54 | 0 | 1 | 11 |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 2 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA……AAABBBBBAAA…AAA

PC

```
8: fd=open("/home/usuario1/fichero1.txt", O_RDWR);
```

devuelve fd=3

datsi © UPM 2024

POLITÉCNICA

## Tablas en memoria

**BCP PA**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PB**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PC**

| fd | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 12 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= ???
UID=101
GID=23

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | 3005 | 2 | 01 |
| 12 | 54 | ~~0~~ 2000 | 1 | 11 |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 2 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

posición 2000

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA……AAABBBBBAAA…AAA

PC

```
9: lseek(fd, 2000, SEEK_CUR);
```

fd=3

datsi © UPM 2024

POLITÉCNICA

## Tablas en memoria

**BCP PA**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PB**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PC**

| fd | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 12 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= ???
UID=101
GID=23

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | 3005 | 2 | 01 |
| 12 | 54 | ~~2000~~-2005 | 1 | 11 |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 2 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

posición 2000

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA……AAACCCCCAAA...AAABBBBBAAA…AAA

PC

```
10: n=write(fd, "CCCCC", 5);
```

fd=3

# APARTADO A: PUNTO DE CONTROL 4; PASO 1

datsi © UPM 2024

POLITÉCNICA

## Tablas en memoria

| BCP PA | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PB | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PC | |
|---|---|
| fd | |
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 12 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| ??? | |
| UID=101 | |
| GID=23 | |

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | ~~3005~~ 1000 | 2 | 01 |
| 12 | 54 | 2005 | 1 | 11 |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 2 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

posición 1000

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA……AAACCCCCAAA...AAABBBBBAAA…AAA

**PB**

```
11: lseek(fd, 1000, SEEK_SET);
```

fd=3

# APARTADO A: PUNTO DE CONTROL 4; FINAL

datsi © UPM 2024

POLITÉCNICA

## Tablas en memoria

**BCP PA**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PB**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PC**

| fd | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 12 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= ???
UID=101
GID=23

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | 1005 | 2 | 01 |
| 12 | 54 | 2005 | 1 | 11 |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 2 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

byte 1000 | byte 2000 | byte 3000 | byte 4999

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA…AAADDDDDAAA…AAACCCCCAAA…AAABBBBBAAA…AAA

**SOLUCIÓN APARTADO C**

**PB**

```
12: n=write(fd, "DDDDD", 5);
```

fd=3

# APARTADO D: PASO 1

datsi © UPM 2024

## Tablas en memoria

**BCP PA**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PB**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PC**

| fd | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 12 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= ???
UID=101
GID=23

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | ~~1005~~ 1000 | 2 | 01 |
| 12 | 54 | 2005 | 1 | 11 |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 2 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

posición 1000

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA…AAADDDDDAAA…AAACCCCCAAA…AAABBBBBAAA…AAA

**PA**

```
13: lseek(fd, 1000, SEEK_SET);
```

fd=3

datsi © UPM 2024

## Tablas en memoria

**BCP PA**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PB**

| fd | |
|---|---|
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= 0022
UID=101
GID=23

**BCP PC**

| fd | |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 12 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |

umask= ???
UID=101
GID=23

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | 1000 | 2 | 01 |
| 12 | 54 | 2005 | 1 | 11 |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 2 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

**ERROR: NO ABIERTO CON PERMISO DE ESCRITURA n=-1**

posición 1000

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA…AAADDDDDAAA…AAACCCCCAAA…AAABBBBBAAA…AAA

**PA**

```
14: n=read(fd, buffer, 5);
```

fd=3

# APARTADO E: PASO 1

datsi © UPM 2024

## Tablas en memoria

| BCP PA | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PB | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PC | |
|---|---|
| fd | |
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 12 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| ??? | |
| UID=101 | |
| GID=23 | |

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | 1000 | 2 | 01 |
| 12 | 54 | ~~2005~~ 1000 | 1 | 11 |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 2 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

posición 1000

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA…AAADDDDDAAA…AAACCCCCAAA…AAABBBBBAAA…AAA

PC

```
13: lseek(fd, 1000, SEEK_SET);
```

fd=3

# APARTADO E: FINAL

datsi © UPM 2024

POLITÉCNICA

## Tablas en memoria

| BCP PA | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PB | |
|---|---|
| fd | |
| 0 | 1 |
| 1 | 2 |
| 2 | 3 |
| 3 | 11 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| 0022 | |
| UID=101 | |
| GID=23 | |

| BCP PC | |
|---|---|
| fd | |
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |
| 3 | 12 |
| 4 | 0 |
| 5 | 0 |
| 6 | 0 |
| 7 | 0 |
| umask= | |
| ??? | |
| UID=101 | |
| GID=23 | |

### Tabla intermedia

| | Nodo-i | Posición | Referen. | rw |
|---|---|---|---|---|
| 11 | 54 | 1000 | 2 | 01 |
| 12 | 54 | ~~1000~~ 1005 | 1 | 11 |
| 13 | | | | |
| 14 | | | | |
| 15 | | | | |
| 16 | | | | |
| 17 | | | | |
| 18 | | | | |

### Tabla de inodos en memoria

| Nodo-i | nopens | agrupaciones | tam |
|---|---|---|---|
| 54 | 2 | 90,91 | 5000 |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

posición
1000 a 1005

**n=5; buffer=DDDDD**

/home/usuario1/fichero1.txt (inodo 54)

rw-r--r-- uid=101 gid=23
tam=5000 agrup=90,91

AAA…AAADDDDDAAA…AAACCCCCAAA…AAABBBBBAAA…AAA

PC

```
14: n=read(fd, buffer, 5);
```

fd=3